

1. Identification of the substance/preparation and of the company/undertaking

Product name	Iloform TDN 81
SDS #	UK-7620, GR-450946
Product Use	Metalworking fluid - neat.
Supplier	Castrol (U.K.) Limited Wakefield House Pipers Way Swindon Wiltshire, SN3 1RE United Kingdom Tel.: +44 (0)1793 512712 Fax.: +44 (0)1793 486083
EMERGENCY TELEPHONE NUMBER	+44 (0)1793 512712

2. Composition/information on ingredients

Highly refined mineral oil and additives. Contains Chlorinated hydrocarbon.

This product does not contain any hazardous ingredients at or above regulated thresholds.

3. Hazards identification

This preparation is not classified as dangerous according to Directive 1999/45/EC as amended and adapted.

Effects and symptoms

Eyes	May cause mild eye irritation.
Skin	Prolonged or repeated contact can defat the skin and lead to irritation and/or dermatitis.
Inhalation	Inhalation of oil mist or vapours at elevated temperatures may cause respiratory irritation.
Ingestion	Ingestion may cause gastrointestinal irritation and diarrhoea.

4. First-aid measures

Eye Contact	In case of contact, immediately flush eyes with a copious amount of water for at least 15 minutes. Get medical attention if irritation occurs.
Skin contact	Wash skin thoroughly with soap and water or use recognised skin cleanser. Remove contaminated clothing and shoes. Wash clothing before reuse. Clean shoes thoroughly before reuse. Get medical attention if irritation develops.
Inhalation	If inhaled, remove to fresh air. Get medical attention if symptoms appear.
Ingestion	Do NOT induce vomiting unless directed to do so by medical personnel. Never give anything by mouth to an unconscious person. If large quantities of this material are swallowed, call a physician immediately.

5. Fire-fighting measures

Extinguishing Media

Suitable	In case of fire, use water fog, foam, dry chemical or CO2 extinguisher or spray.
Not Suitable	Do not use water jet.

Hazardous decomposition products These products are carbon oxides (CO, CO2), halogenated compounds, hydrogen chloride.

Unusual fire/explosion Hazards This material is not explosive as defined by established regulatory criteria.

Special fire-fighting procedures None identified.

Protection of fire-fighters Fire fighters should wear self-contained positive pressure breathing apparatus (SCBA) and full turnout gear.

6. Accidental release measures

Personal Precautions	Immediately contact emergency personnel. Keep unnecessary personnel away. Use suitable protective equipment (Section 8). Follow all fire fighting procedures (Section 5).
Environmental precautions and cleanup methods	If emergency personnel are unavailable, contain spilled material. For small spills add absorbent (soil may be used in the absence of other suitable materials) scoop up material and place in a sealed, liquid-proof container for disposal. For large spills dike spilled material or otherwise contain material to ensure runoff does not reach a waterway. Place spilled material in an appropriate container for disposal. Minimize contact of spilled material with soils to prevent runoff to surface waterways. See Section 13 for Waste Disposal Information.
Personal Protection in Case of a Large Spill	Splash goggles. Full suit. Boots. Gloves. Suggested protective clothing might not be sufficient; consult a specialist BEFORE handling this product.

7. Handling and storage

Handling	Wash thoroughly after handling. Concentrations of mist, fumes and vapours in enclosed spaces may result in the formation of explosive atmospheres. Excessive splashing, agitation or heating must be avoided.
Storage	Keep container tightly closed. Keep container in a cool, well-ventilated area.

8. Exposure controls/personal protection

Ingredient Name	Occupational Exposure Limits
Highly refined mineral oil	EH40-OES (United Kingdom (UK)). TWA: 5 mg/m ³ 8 hour(s). Form: Oil mist, mineral STEL: 10 mg/m ³ 15 minute(s). Form: Oil mist, mineral
Control Measures	Provide exhaust ventilation or other engineering controls to keep the airborne concentrations of vapours below their respective occupational exposure limits. Local exhaust ventilation is preferred because it prevents contaminant dispersion into the work area by controlling it at its source.
Hygiene measures	Wash hands after handling compounds and before eating, smoking, using lavatory, and at the end of day.
Personal protective equipment	
Respiratory system	None required; however, use of adequate ventilation is good industrial practice.
Skin and body	Wear appropriate clothing to avoid prolonged skin contact.
Hands	Wear protective gloves if prolonged or repeated contact is likely. Consult your local supplier of personal protective clothing regarding the choice of material.
Eyes	Safety glasses with side shields.

9. Physical and chemical properties

Autoignition temperature	>250 °C
Flash point	>170 °C (CLOSED CUP)
Colour	Amber.
Odour	Mild
Odour threshold	Not available.
Physical state	Liquid.
Boiling point / range	>250 °C
Density	>1 g/cm ³ at 20 °C
Vapour pressure	<0.075 mmHg
Solubility	Insoluble in cold water.
Viscosity	kinematic at 40°C: 170 cSt

10. Stability and reactivity

Incompatibility with Various Substances	Reactive with oxidizing agents.
Hazardous Polymerization	Will not occur.

Product Name Iloform TDN 81	Product Code UK-7620	Page: 2/4
Version 1	Date of issue 6 December 2002	Format United Kingdom (UK)
	(BUILD 5.0)	Language (ENGLISH)

11. Toxicological information

Chronic toxicity

Carcinogenic effects

No component of this product at levels greater than 0.1% is identified as a carcinogen by ACGIH, the International Agency for Research on Cancer (IARC) or the European Commission (EC).

12. Ecological information

Persistence/degradability

Not readily biodegradable.

Mobility

Nonvolatile. Viscous liquid. Insoluble in water.

Environmental hazards

Not classified as dangerous.

13. Disposal considerations

Disposal Consideration / Waste information

Avoid contact of spilled material and runoff with soil and surface waterways. Consult an environmental professional to determine if local, regional or national regulations would classify spilled or contaminated materials as hazardous waste. Use only approved transporters, recyclers, treatment, storage or disposal facilities.

Hazardous Waste

This product is listed as Hazardous by the EU Directive on hazardous waste. Dispose of according to all national and local applicable regulations.

14. Transport information

Not classified as hazardous for transport (ADR, RID, UN , IMO, IATA/ICAO).

15. Regulatory information

Label Requirements

Risk Phrases

This product is not classified according to the EU regulations.

EU Regulations

Classification and labelling have been performed according to EU directives 1999/45/EC and 67/548/EEC as amended and adapted.

Other Regulations

Inventories

AUSTRALIAN INVENTORY (AICS): Listed on inventory.

CANADA INVENTORY (DSL): Listed on inventory.

CHINA INVENTORY (IECS): Listed on inventory.

EC INVENTORY (EINECS): Listed on inventory.

JAPAN INVENTORY (ENCS): Not determined.

KOREA INVENTORY (ECL): Listed on inventory.

PHILIPPINE INVENTORY (PICCS): Listed on inventory.

US INVENTORY (TSCA): Not determined.

Additional Warning Phrases

Safety data sheet available for professional user on request.

16. Other information

HISTORY

Date of issue

06/12/2002.

Date of previous issue

No Previous Validation.

Prepared by

Product Stewardship

Notice to Reader

The data and advice given apply when the product is sold for the stated application or applications. The product is not sold as suitable for any other application. Use of the product for applications other than as stated in this sheet may give rise to risks not mentioned in this sheet. You should not use the product other than for the stated application or applications without seeking advice from us.

If you have purchased the product for supply to a third party for use at work, it is your duty to take all necessary steps to secure that any person handling or using the product is provided with the information in this sheet.

If you are an employer, it is your duty to tell your employees and others who may be affected of any hazards described in this sheet and of any precautions which should be taken.

Further copies of this Safety Data Sheet may be obtained from Castrol.

Product Name Iloform TDN 81		Product Code UK-7620	Page: 4/4
Version 1	Date of issue 6 December 2002	Format United Kingdom (UK) (BUILD 5.0)	Language (ENGLISH)